The Wedge Zome: How to Cut and Build It

A nine-sided zome of 36 rhombus panels in 4 rows, 13.8 ft across and 8.7 ft tall, every panel side 35.43 in. It is the Complete Edition's zome, framed with wedges split 8 ways from a 12 in log – the same wedges that frame the dome.

The idea in one paragraph

Every panel is four wedge sticks in a pinwheel, framed flat on a jig, with the stick's flat back on the **outside** and its point toward the inside. A wedge's sides lean 22.5° away from square, so where two panels fold together their backs close tight along the outside seam and the inside opens into a V of 17.2° to 30.8° – the wedge's 45° point less the fold, which runs from 14.2° to 27.8°. No block fills anything. The inside V is a channel for wire, air or water, capped or left open.

What it takes

	this method	the other method
sticks	144	144
wedges, 72 in long	72	144
log sections, split 8 ways	9	18
sticks from one wedge	2	2

Every stick, and how to cut it

Lay the stick **back down** on the saw table with the edge it runs along against the fence, a 22.5° shim under the fence side so the sloping face bears flat – the shim is a wedge split 16 ways. Mitre is the blade's swing on the table, bevel its tilt.

stick	row	how many	length, outside edge	tail: mitre, bevel	head: mitre, bevel
R1A	1	18	27.31 in	57.9° L, 22.5°	57.9° L, 22.5°
R1B	1	18	27.31 in	57.9° R, 22.5°	57.9° R, 22.5°
R2A	2	18	30.57 in	27.3° L, 22.5°	27.3° L, 22.5°
R2B	2	18	30.57 in	27.3° R, 22.5°	27.3° R, 22.5°
R3A	3	18	31.11 in	1.0° L, 22.5°	1.0° L, 22.5°
R3B	3	18	31.11 in	1.0° R, 22.5°	1.0° R, 22.5°
R4A	4	18	30.95 in	15.6° R, 22.5°	15.6° R, 22.5°
R4B	4	18	30.95 in	15.6° L, 22.5°	15.6° L, 22.5°

Every end of every stick takes the same bevel, 22.5° – half the wedge's point. Set it once. Only the mitre changes, once per row, and the two stick types in a row are mirror images: the same mitre swung the other way.

The rows

Row 1 is the crown, the 9 panels round the point; row 4 stands on the sill.

row	panels	sharp corner	largest fold	inside V
1	9	32.1°	24.2°	20.8°–30.8°
2	9	62.7°	19.3°	25.7°–30.8°
3	9	89.0°	27.8°	17.2°–25.7°
4	9	74.4°	27.8°	17.2°–17.2°

Order of work

1. **Buck and split.** Buck 9 sections, 72 in long; rip each 8 ways through the pith with the chainsaw, never alone. Trim the backs flat.
2. **Cut the sticks.** 2 sticks from each wedge. Set the bevel once; cut every stick of a row before changing the mitre.
3. **Build the jig, one per row.** Screw stops to a sheet at the row's rhombus outline, full size, from the jig drawing for that row.

4. **Frame the panels.** Four sticks in a pinwheel: each runs to the corner ahead of it and its tail butts the side of the stick behind. Glue and screw at each butt, in the thick part, pilot holes always.
5. **The sill and the rim posts.** The rim zigzags: 18 corners, 9 of them 20.8 in above the sill. Stand a post under each high corner – two quarters back to back make a square post 5.62 in on a side.
6. **Raise, bottom row up.** Fasten panels to each other through the outside seam. Brace the top rows: they sag until the ring closes.
7. **Close the seams.** The outside seams are closed by the backs: seal them with the seam cap. The inside V is a channel – run wire or air in it, then cap it.

Before anyone lives in it

This is a method, not a rated structure. Have a licensed engineer check the frame for wind and snow, and the fastening schedule, before it is occupied.